LESSON 8.6

Solve and Graph Inequalities

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can solve an inequality and graph its solution set on a number line.

WHY THIS MATTERS Detective Kim knows that the suspect withdrew more than $50 from an ATM. The amount plus a $3 fee totaled more than $53. She writes x + 3 > 53 and needs to solve for x, then graph the solution set to share with the bank. How can she find and display all possible withdrawal amounts?

REMEMBER

Solve and Graph Inequalities

Solving & Graphing One-Step Inequalities I solve an inequality the same way as an equation: use inverse operations to get the variable alone. But the answer is a range of values, so I show all of them on a number line.

1. Isolate the variable using inverse operations on both sides (same as equations).
2. Write the simplified inequality with the variable on the left (e.g. x > 5).
3. Graph the solution set on a number line with open/closed circle and arrow shading

Words: **Solve *(Resolver)*** **Graph *(Graficar)*** **Solution *(Solución)*** **Substitute *(Sustituir)*** **Solution set *(Conjunto solución)***

1 **Circle the letter of the best answer. Show how you know.**

Which graph represents x > 8?

(A) Closed circle at 8, shaded right
(B) Open circle at 8, shaded right
(C) Open circle at 8, shaded left
(D) Closed circle at 8, shaded left

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

Which graph represents $x \leq 10$?

(A) Open circle at 10, shaded left

(B) Closed circle at 10, shaded left

(C) Closed circle at 10, shaded right

(D) Open circle at 10, shaded right

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Which graph represents $x < 5$?

(A) Closed circle at 5, shaded left

(B) Open circle at 5, shaded right

(C) Open circle at 5, shaded left

(D) Closed circle at 5, shaded right

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

A detective needs at least \$75 for a surveillance camera. She has already saved \$48. The inequality $48 + s \geq 75$ represents how much more money s she needs. Solve $48 + s \geq 75$.

(A) $s \geq 27$

(B) $s \leq 27$

(C) $s \geq 75$

(D) $s \geq 123$

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

A detective needs at least $75 for a surveillance camera. She has already saved $48. The inequality 48 + s ≥ 75 represents how much more money s she needs. Is exactly $27 enough?

(A) No — she needs more than 27
(B) Yes — 'at least' includes 27 exactly
(C) No — she needs 75
(D) Only if she saves more

SHOW YOUR WORK

6 **Circle the letter of the best answer. Show how you know.**

A detective needs at least $75 for a surveillance camera. She has already saved $48. The inequality 48 + s ≥ 75 represents how much more money s she needs. How is s ≥ 27 graphed?

(A) Closed circle at 27, shaded to the right
(B) Open circle at 27, shaded to the right
(C) Closed circle at 27, shaded to the left
(D) Open circle at 27, shaded to the left

SHOW YOUR WORK

EXPLAIN YOUR THINKING

How is solving an inequality similar to and different from solving an equation?

SENTENCE STARTER Solving an inequality is like solving an equation because I still use ___, but it is different because the answer is ___ instead of ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

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LEARNING TARGET I can solve an inequality and graph its solution set on a number line.

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PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Detective, sort each suspect's badge number into the right vault door for the rule x ≥ 4. A value is a solution if it makes x ≥ 4 true.

Groups: Solution of x ≥ 4 | NOT a solution of x ≥ 4

_____ 9

_____ 5

_____ 4

_____ 3

_____ 2

_____ 0

2 **Write the letter of the matching item on each line.**

Match each inequality to the equivalent inequality written with the number first. Both inequalities in a pair must have the same solutions.

_____ x > 10

_____ x ≤ 6

_____ x ≥ 8

_____ x < 5

(A) 10 < x

(B) 6 ≥ x

(C) 8 ≤ x

(D) 5 > x

3 **Circle the letter of the best answer. Show how you know.**

Final gate — describe the solution set. The vault rule is x > 7. What does the solution set mean?

(A) All numbers greater than 7
(B) Only the number 7
(C) All numbers less than 7
(D) Every number, including 7

SHOW YOUR WORK

PART 2 Apply it

real-world problems

4 **Circle the letter of the best answer. Show how you know.**

Clue 1: Which value is a solution of x > 7?

(A) 3
(B) 5
(C) 7
(D) 9

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Clue 2: Is 5 a solution of x ≤ 5?

(A) Yes, because 5 ≤ 5 is true
(B) No, because 5 is not less than 5
(C) No, because ≤ means only smaller numbers
(D) Yes, but only because 5 is odd

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 Find the mistake. Explain it, then show the correct work.

Level 2 Extension — Catch the Boundary Mistake

1. **Vault rule** x > 7
2. **Test the boundary value** Substitute x = 7: 7 > 7
3. **Rookie's conclusion** 7 > 7 is true, so 7 IS a solution and opens the gate.
4. **Mark the solution set** The detective adds 7 to the list of solutions.

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

How is solving an inequality similar to and different from solving an equation?

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